StoryTutor-MM

Research Validation Plan

From completed product to defensible research contributions

This revised guide treats the end-to-end product workflow as already implemented, as shown in the supplied workflow infographic and the 77-second demo. The research task is therefore not to build these features again. It is to establish, with carefully designed experiments, how reliably they work and what the verification and retrieval methods contribute.

What is demonstrated in the supplied materials	What remains a research question
Question-answer chat; class/subject/language selection; source-page references; read-aloud controls; generated short lesson videos; fact-check labels; lesson/video library and prior-answer flow.	Accuracy and reliability under representative test conditions; whether citations point to truly supporting pages; whether flagged video claims are truly unsupported; robustness across language, grade, subject and diagram claims.
The project plan also describes hybrid retrieval, query rewrite rescue, an evidence gate, guardrails, a 268-question evaluation set and per-claim checks for narration and diagram relationships.	Independent baselines, human-labelled test data, ablations, confidence intervals, failure analysis, licence-safe dataset release and a defensible novelty comparison with prior work.

Central recommendation

Lead with claim-level verification of curriculum-grounded explainer videos. Treat the multilingual RAG benchmark as a strong supporting study. Do not claim numerical accuracy, learning gains, or priority over previous work until measured.

Prepared from the supplied RESEARCH_PLAN.md, PRODUCT_PLAN.md, workflow image and demo video. Product capabilities here are taken from these materials and visible demo behavior; this report does not independently audit the codebase or claim that accuracy metrics have already been measured.

1. Reframe the paper around what is already built

Your correction changes the research plan materially. The system is no longer best described as a proposed tutor architecture. It is an implemented, end-to-end system whose core components can now be evaluated as a research object.

Implemented capability reported or shown	Do not claim yet without measurement
Grounded Q&A; with visible source/page attribution	That every answer is correct or every citation genuinely entails the answer.
English, Hindi and Marathi user flows and read-aloud controls	Equivalent quality across languages, accents, scripts or code-switched questions.
Generated short lesson videos and visible fact-check status	That every spoken statement, image, arrow, label or diagram relation is supported.
Hybrid dense + BM25 retrieval, reranking and one rewrite rescue	That the combination outperforms dense-only, BM25-only or no-rewrite baselines.
268-question golden set and evaluation harness described in the plan	That the set is representative, independent, balanced or ready for public release.
7,086 curriculum chunks reported in the product plan	That corpus size alone implies coverage, completeness, accuracy or rights to redistribute.

The distinction to preserve in a paper

- **Implementation claim:** the product contains a claim-verification pipeline and exposes its verdict to users.
- **Empirical claim:** the pipeline detects unsupported video claims with a measured precision/recall/F1 on held-out human-labelled examples.
- **Novelty claim:** this formulation, dataset or method fills a gap relative to prior methods, supported by a systematic review and direct comparisons.

A demo is evidence of the workflow and feasibility. It is not a substitute for an independently annotated test set.

2. Lead research direction: verify generated educational videos

Suggested working title

Evidence-Grounded Verification of Generated Educational Explainer Videos in Multilingual, Curriculum-Bounded Systems

Main research question

Can an evidence-linked verifier reliably identify unsupported claims in automatically generated educational videos, covering both narrated factual statements and diagram-relation claims, across English, Hindi and Marathi?

What makes your actual system worth investigating

- It does not stop at evaluating the text answer: it checks claims in a generated video.
- The verification unit includes narration claims [C...] and diagram-edge claims [D...], such as whether a labelled arrow or relationship is supported by the source.
- The pipeline described in your plan uses an LLM judge, embedding-similarity checks and a lexical fallback, with verdicts recorded in a sidecar report and surfaced in the UI.
- The evidence source is a bounded curriculum corpus with page-level attribution, rather than unconstrained general-web grounding.

These are plausible components of a contribution, not proof of worldwide priority. A novelty claim such as “the first system” should only be made after a systematic literature review across video factuality, multimodal claim verification, grounded generation and educational technology.

Proposed contribution package

Contribution	What the paper should deliver
Method	A precise formalization of claim extraction, evidence linking, verdicts, uncertainty/abstention and fallback behavior, based on the already implemented pipeline.
Dataset / protocol	Human-labelled claim-evidence-verdict examples, including narration and diagram relations, with language, subject, grade and evidence-page labels where lawful.
Evaluation	Direct comparison of cascade verifier against LLM-only, embedding-only, lexical-only and no-verifier baselines.
Analysis	Errors by language, claim type and subject; judge/embedding disagreement; threshold sensitivity; evidence quality; false assurances.

3. Metrics that make the claim measurable

For verification, define the positive class as **an unsupported claim**. This makes recall particularly important: it measures how many unsupported claims the system actually catches.

Metric	Definition / interpretation
Precision	TP / (TP + FP). Of the claims flagged unsupported, how many truly are unsupported? Low precision means too many safe claims are wrongly flagged.
Recall / sensitivity	TP / (TP + FN). Of all genuinely unsupported claims, how many did the verifier catch? Low recall means unsupported content slips through.
F1 score	2 × (precision × recall) / (precision + recall). A single summary of precision/recall balance; still report both separately.
Specificity	TN / (TN + FP). How well does the verifier leave supported claims unflagged?
False-negative rate	FN / (TP + FN). The fraction of unsupported claims the system misses. Especially important for factual safety.
Balanced accuracy	Average of recall and specificity. Useful when supported and unsupported claims are imbalanced.
Macro-F1	Compute F1 within each language or claim class, then average equally. Stops the largest language from dominating the headline.
Inter-annotator agreement	Cohen’s kappa or Krippendorff's alpha for independent human labels; report raw agreement too. Shows label reliability.

Confusion matrix for unsupported-claim detection

	Human says unsupported	Human says supported
Verifier flags unsupported	TP: correctly caught	FP: false alarm
Verifier says supported	FN: missed unsupported claim	TN: correctly left alone

Do not report accuracy alone. If most generated claims are supported, a system that almost always says “supported” can show high accuracy while missing the dangerous minority of unsupported claims. Use 95% confidence intervals, preferably paired bootstrap intervals for model comparisons.

4. Build the annotation set around the real failure mode

The project plan says a 268-question golden set already exists for RAG/chat evaluation. That is useful. For the video-verification paper, create a separate claim-level evaluation set: questions and answers are not the same unit as generated narration claims and diagram relations.

Suggested annotation unit

Field	Purpose
video_id / claim_id	Stable identifier for the generated clip and each extracted claim.
language, grade, subject, chapter	Stratification and error analysis; freeze values at generation time.
claim text or relation	Narration proposition or diagram-edge proposition; label the type explicitly.
source page / evidence span	The passage, page or evidence IDs the verifier used; keep traceability to the book.
human label	Supported / contradicted / not supported / ambiguous, with a written annotation policy.
system decision + confidence	Preserve the raw LLM judgement, embedding score, lexical signal, cascade verdict and fallback path.
error category	Extraction error, missing evidence, wrong evidence, overconfident verdict, diagram mismatch, translation or ambiguity.

A practical pilot design

- Start with a manageable, stratified sample across all three languages, the three grade levels, Science/Social Science, and both narration and diagram-edge claims. A planning target could be 500-1,000 annotated claims, then adjust based on label prevalence and confidence-interval width.
- Have two independent annotators review each sampled claim against the relevant source. Prefer a teacher or subject expert for at least a validation subset; adjudicate disagreements and report agreement.
- Separate development data from the held-out test data by chapter, concept or generation template where feasible. Do not let near-duplicate question/video variants leak across splits.
- Tune thresholds on a development set only. Lock the test set before the final comparisons.

The sample size above is a planning recommendation, not a guarantee of statistical power. If the data are sparse in one language or in diagram claims, report that limitation instead of overgeneralizing.

5. Baselines and ablations: show what causes the gain

Because your verifier is already implemented, the research work is to compare its components under the same held-out examples and judging protocol.

Variant	Question answered
No verifier	How often would generated videos contain unsupported claims if the generator output were accepted as-is?
LLM judge only	How well does the judgement model do without the embedding and lexical fallback?
Embedding similarity only	Can similarity to retrieved evidence reliably detect unsupported claims?
Lexical / keyword baseline only	How much can simple overlap detect, and where does it fail?
Current cascade: judge → embedding → keyword fallback	Does the full system improve the precision/recall trade-off?
Cascade without diagram-edge claims	What added value or added error comes from checking diagram relationships?
Threshold sensitivity	Do results change materially across the current 0.55/0.75 embedding thresholds and neighbouring values?
Language-specific slices	Does the verifier degrade on Hindi/Marathi, romanized queries or mixed-language narration?

Important controls

- Use the same claim set, evidence context, and source material for each baseline.
- Use an independent judge or human labels for the final outcome; avoid using the same model both to generate and conclusively score its own output.
- Report outcomes separately for narration claims and diagram claims before aggregating.
- Report runtime and cost per video as secondary systems metrics; extra verification is useful only if the quality benefit is worth its overhead.

The key result is not “our pipeline has several components.” It is a measured difference: for example, whether the cascade catches more unsupported claims at an acceptable false-alarm rate, and which component contributes that difference.

6. Supporting study: multilingual RAG and page attribution

The second strongest paper direction is to evaluate whether an answer is grounded in the correct approved textbook page, including cross-lingual cases such as a Marathi question answered using an English-medium source page.

Task	Recommended metrics	Interpretation
Retrieval	Recall@1 / @5 / @10, MRR, nDCG@k	Did retrieval surface the gold page/evidence, and how high was it ranked?
Page attribution	Citation precision, citation recall, exact page accuracy	Are cited pages correct and do they support the answer?
Answer faithfulness	Human-labelled supported-claim proportion; groundedness rubric	What fraction of answer claims are supported by the retrieved source?
Abstention/refusal	Precision, recall, false-refusal rate, missed-unanswerable rate	Does the system refuse when evidence is insufficient without refusing answerable questions?
Language robustness	Macro-F1 and metrics sliced by input language, source language and code-switch style	Does the system work across English/Hindi/Marathi and cross-lingual retrieval?
Teaching quality	Blind teacher rubric: correctness, level fit, clarity, useful example, unsupported additions	Is the answer correct and usable for the intended class level?
Operations	p50/p95 latency, first-token latency, failure rate, cost per answer/video	Can the approach run reliably, at realistic loads and cost?

RAG baselines and ablations

- Dense retrieval only; BM25 only; hybrid dense + BM25 with reciprocal-rank fusion; hybrid + cross-encoder reranking; and hybrid + query rewrite rescue.
- Compare against a closed-book language model for answer correctness and unsupported additions, but do not mix the retrieval and generation questions into one opaque score.
- Use chapter- or concept-held-out splits when creating evaluation partitions, not only random question splits. Near-duplicate curriculum questions can otherwise make results look better than true generalization.

The existing 268-question set is a seed, not automatic proof of benchmark readiness. Document how questions were sourced, answerability labels, page ground truth, language coverage, class/subject balance and any annotation disagreements.

7. Statistical reporting and result tables

Predefine the primary outcomes before running the final evaluation. For the lead paper, a sensible primary outcome is unsupported-claim recall, paired with precision/F1 and a false-negative analysis. For RAG, predefine page-level Recall@5 and exact citation correctness.

Practice	Why it matters
95% confidence intervals	Show the plausible uncertainty around each score; avoid presenting a small test set as a precise universal truth.
Paired comparison	Evaluate every method on the same examples; use paired bootstrap intervals for score differences, and a suitable paired test such as McNemar for binary errors where applicable.
Stratified reporting	Break out language, grade, subject, claim type and answerability; give macro averages and per-slice sample counts.
Threshold discipline	Tune thresholds on development data, then freeze before the held-out test. Publish sensitivity analysis separately.
Error analysis	Manually categorize false negatives, false positives, evidence misses, extraction errors, translation issues and diagram mismatches.
Version control	Freeze model versions, prompts, corpus snapshot, embedding model, threshold configuration and evaluation code for every paper run.
Reproducibility	Commit scripts, config files, raw aggregate reports, environment and seed information; document any API-backed components.

Example results table (fill only after experiments)

Method	Precision	Recall	F1	False-negative rate
No verifier	TBD	TBD	TBD	TBD
LLM judge only	TBD	TBD	TBD	TBD
Embedding only	TBD	TBD	TBD	TBD
Lexical only	TBD	TBD	TBD	TBD
Current cascade	TBD	TBD	TBD	TBD

Never fabricate these numbers, extrapolate them from a demo, or turn evaluator output into “ground truth” without validation. Use “we evaluate” until you have results; use “we achieve X” only after the experiment is complete.

8. What you can claim now - and what needs evidence

Claim category	Defensible wording today	Requires experiments
System capability	“We implemented a multilingual curriculum-grounded tutor with page-linked answers, read-aloud output, short generated lessons and a visible video-checking workflow.”	Not a performance claim; still document code and demo versions.
Verifier design	“The pipeline checks narration claims and diagram-relation claims against retrieved curriculum evidence using a multi-signal cascade.”	Human-labelled precision/recall, agreement, failure modes and baseline comparisons.
Grounding	“The product is designed to answer from an approved corpus and show source pages.”	“Only answers from the book” requires testing of unsupported additions, retrieval misses and citation correctness.
Multilingual quality	“The interface and generation workflow support English, Hindi and Marathi.”	Equivalent accuracy or quality across languages requires per-language measurements.
Learning impact	No learning-gain claim yet.	Pre/post study, appropriate consent and ethics review, control condition, effect size and confidence interval.
Novelty / firstness	“We study claim-level verification of narration and diagram relations for multilingual curriculum-grounded lesson videos.”	“First” or “state of the art” requires systematic literature review and direct comparative evidence.

A more defensible novelty statement

> **“We present an evidence-grounded evaluation framework for generated educational explainer videos that treats narration claims and diagram relations as separate verification units, links decisions to curriculum-page evidence, and evaluates a multi-signal verifier across English, Hindi and Marathi.”**

Use this as a draft positioning statement, not a proven novelty claim. Narrow it further if prior work already covers parts of the method. Your novelty may lie in the combination, formalization, dataset, language setting or empirical findings rather than in any one idea alone.

9. Related studies to keep in reserve

Study	Why it is useful	Priority
NCERT-Ground: trilingual groundedness and page attribution	A reproducible evaluation benchmark for page correctness, refusal, cross-lingual retrieval and grounded answers. Must solve source-text redistribution/licensing before release.	High; complementary paper
Code-switched / romanized retrieval	Measure native-script vs romanized vs code-switched queries and whether query rewrite rescues retrieval or introduces drift.	High; once query labels are ready
Devanagari extraction quality	Validate the no-reference text damage/repair metric against human inspection and downstream retrieval degradation.	Efficient, data-centric paper
Small teaching model quality-cost frontier	Compare 1.7B/8B/14B/24B variants using blinded teacher ratings, groundedness, latency and cost.	Useful systems follow-up
Learning gain and delayed retention	Compare StoryTutor with textbook-only learning and potentially generic AI answers. This measures educational impact, not just technical quality.	High impact; later with school/ethics approvals

Content licensing and release

- Do not publicly redistribute NCERT PDFs, extracted textbook paragraphs or chunks unless you have the rights to do so.
- A safer benchmark design may release questions, labels, page identifiers, offsets or evidence IDs plus code that lets authorized users ingest their own copies - but verify whether each proposed field is itself permissible to release.
- Maintain source/license records for model weights, code, fonts, datasets and generated assets. “Hosted on Hugging Face” or “downloaded from an official website” does not by itself establish open-source or commercial reuse rights.
- For a learner study, obtain appropriate institutional ethics review, school authorization and required parental consent before collecting student-level outcomes. Use minimal, anonymized data and a pre-specified analysis plan.

10. Research execution plan - no feature rebuild required

Phase	Actions	Deliverable
Days 1-3: freeze scope	Choose the lead question; define unsupported-claim positive class; record system, model, prompt and corpus versions; review closest literature.	One-page protocol and novelty map.
Days 4-8: sampling and labels	Sample clips/claims across grade, language, subject and claim type; write annotation guide; run independent labels and resolve disagreements.	Claim-level gold set with annotation agreement.
Days 9-13: reproducible runs	Evaluate no-verifier, LLM-only, embedding-only, lexical-only and current cascade using a frozen test set.	Raw predictions and machine-readable metrics.
Days 14-17: ablations	Threshold sensitivity; narration vs diagram claims; language and subject slices; latency/cost.	Ablation and error analysis tables.
Days 18-22: statistics	Confidence intervals, paired comparisons, error taxonomy and robustness checks.	Final tables and figures.
Days 23-30: writing	Draft method, dataset protocol, related work, results, limitations, ethics and release plan.	Journal/conference manuscript draft.

Proposed paper outline

- Introduction: the risk of unsupported claims in generated educational videos.
- Related work: grounded generation, factuality verification, video/multimodal verification, educational AI, multilingual NLP.
- System and method: claim extraction, evidence linking, verifier cascade, verdict and fallback behavior.
- Dataset and annotation: sampling, labels, evidence policy, split design, agreement and rights.
- Experiments: baselines, metrics, statistical analysis and ablations.
- Results and error analysis: report both successes and failure modes; include language/diagram slices.
- Limitations, safety, licensing, reproducibility and conclusion.

11. Final strategic recommendation

Treat the completed product as your experimental platform. Your next meaningful work is to freeze versions, annotate a reliable gold set, compare the existing verifier with simpler baselines, and report where it succeeds and fails. That is how the project moves from a compelling demo to a defensible research contribution.

Suggested priority order

Order	Focus	Reason
1	Video claim verification	Most distinctive feature bundle in your current research plan: narration + diagram relations + curriculum evidence + trilingual setting.
2	Trilingual RAG/citation benchmark	Uses existing 268-question seed and retrieval harness; gives the paper reproducible grounding evidence.
3	Code-switched retrieval or Devanagari extraction	Potentially efficient follow-on studies grounded in existing assets.
4	Learning-gain study	High educational impact, but requires school partnership, study design, ethics and consent.

The result you are aiming for

A paper that makes a narrow, falsifiable claim; evaluates it with human-checked ground truth; compares the existing method against clear baselines; reports uncertainty and failures; and releases what can legally be shared.

Source basis: user-supplied RESEARCH_PLAN.md, PRODUCT_PLAN.md, the K12 workflow image and the StoryTutor demo video. Features are treated as implemented where described or visibly demonstrated. Quantitative results, comparative superiority, learning gains and novelty priority are not asserted unless supported by future experiments. This report is research planning guidance, not legal advice.